

Post-diagnosis dietary factors and survival after invasive breast cancer

Little is known about the effects of diet after breast cancer diagnosis on survival. We prospectively examined the relation between post-diagnosis dietary factors and breast cancer and all-cause survival in women with a history of invasive breast cancer diagnosed between 1987 and 1999 (at ages 20–79 years). Diet after breast cancer diagnosis was measured using a 126-item food frequency questionnaire. Among 4,441 women without a history of breast cancer recurrence prior to completing the questionnaire, 137 subsequently died from breast cancer within 7 years of enrollment. Hazard ratios (HR) and 95% confidence intervals (CI) were estimated for intake of macronutrients as well as selected micronutrients and food groups from Cox proportional hazards regression models. After adjustment for factors at diagnosis (age, state of residence, menopausal status, smoking, breast cancer stage, alcohol, history of hormone replacement therapy), interval between diagnosis and diet assessment, and at follow-up (energy intake, breast cancer treatment, body mass index, and physical activity), women in the highest compared to lowest quintile of intake of saturated fat and trans fat had a significantly higher risk of dying from any cause (HR = 1.41, 95% CI = 1.06 to 1.87, P-trend = 0.03) for saturated fat; (HR = 1.78, 95% CI = 1.35 to 2.32, P-trend = 0.01) for trans fat intake. Associations were similar, though did not achieve statistical significance, for breast cancer survival. This study suggests that lower intake of saturated and trans fat in the post-diagnosis diet is associated with improved survival after breast cancer diagnosis.